



E-Commerce Sales & Customer Behaviour Analytics

using Big Data — Apache Spark · MongoDB · Machine Learning

Presented by

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Guide: <Guide Name>

LY Innovation-Based Major Project-B · Dept. of Computer Engineering

K. J. Somaiya Institute of Technology, Sion, Mumbai · 2026-27



Agenda

01 Problem & objectives

02 Literature & research gaps

03 Technology stack & architecture

04 Data pipeline & data quality

05 Machine-learning models & results

06 Live dashboard walkthrough

07 Business insights

08 Conclusion & future scope

The problem: data-rich, insight-poor online retailers



Too big & too varied

Orders, catalogues, profiles and 1 M+ click-stream events overwhelm spreadsheets and single servers.



Messy data

Duplicates, "upi" vs "UPI", missing cities and invalid quantities silently corrupt reports.



Fragmented tools

Sales, funnel, segmentation and forecasts live in separate tools with no single customer view.



Reactive retention

Valuable customers are noticed only after they have left, with no warning and no explanation.



Generic merchandising

Everyone sees the same best-sellers; bundles are designed by intuition.



No foresight

Plans ignore festive-sale effects; outages and bulk orders go unnoticed.

Aim & objectives

Build an end-to-end platform that uses Apache Spark, MongoDB and machine learning to analyse e-commerce sales and customer behaviour and deliver predictive, explainable insights through an interactive dashboard.

Generate

Realistic 2-year dataset with seasonality, churn & dirty data

Engineer

Bronze → silver → gold Spark pipeline with a data-quality report

Analyse

KPIs, geo, funnels, sessions & cohort retention (Spark SQL)

Store

Query-shaped MongoDB documents written in parallel

Predict

Segments, churn, recommendations, bundles, forecast, anomalies

Deliver

REST API + 11-page dashboard, simulator & CSV export

Why is this a Big Data problem?

1.35 M

click-stream events (JSON-lines,
24 monthly files)

74,348

orders with 1.45 lakh line items

10,000

customers across 28 Indian cities

315 MB

raw data per run (medium
scale); 790 MB at large scale

Volume

≈1.4 million records per run; scales to 3.4 M+ events without code changes.

Variety

Structured CSV tables + semi-structured JSON logs + product catalogue with nested fields.

Velocity

Every visit produces 4–7 events; daily batch runs, with a streaming path (Kafka) as future work.

Literature review & research gaps

Key references (25 reviewed)

Spark / RDD — Zaharia et al. 2012, 2016

In-memory distributed processing

MLlib — Meng et al. 2016

Scalable ML pipelines

RFM + k-means — Chen et al. 2012

Customer segmentation

Churn benchmarks — Verbeke et al. 2012

Classifier comparison

Implicit ALS — Hu, Koren & Volinsky 2008

Recommendations from clicks

FP-Growth — Han et al. 2000

Frequent itemsets

Prophet — Taylor & Letham 2018

Holiday-aware forecasting

MAD outliers — Leys et al. 2013

Robust anomaly detection

Gap → how ShopSense addresses it

Isolated solutions → One pipeline feeds all models + Customer 360

Temporal leakage in churn → Strict observation / label windows

Explicit-rating assumption → Implicit views, carts & purchases

Data quality ignored → Measured cleaning with a DQ report

Naïve forecasts & alarms → Sale-calendar features, robust residual z-score

Metrics, not actions → Playbooks, explanations, simulator, CSV export

Technology stack



Python 3.11

Pipeline, ML & API code



Apache Spark 3.5

PySpark DataFrames, Spark SQL



Spark MLlib

K-Means · LR · RF · GBT · ALS · FP-Growth



MongoDB 8

Document store, aggregation pipelines



FastAPI

29 REST endpoints + Swagger docs



ECharts + JS

11-page responsive dashboard

System architecture



1 Sources

CSV + JSON-lines landing zone

2 Spark

bronze → silver → gold, MLlib

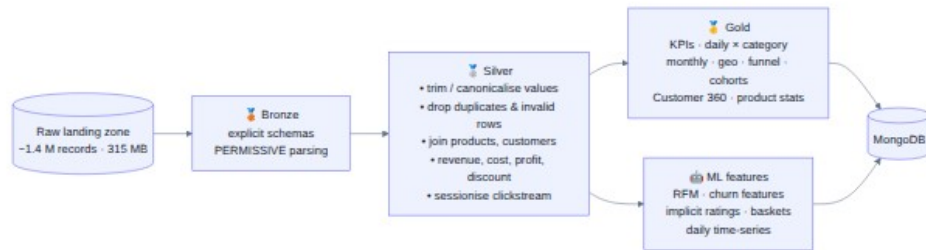
3 MongoDB

query-shaped gold documents

4 Serving

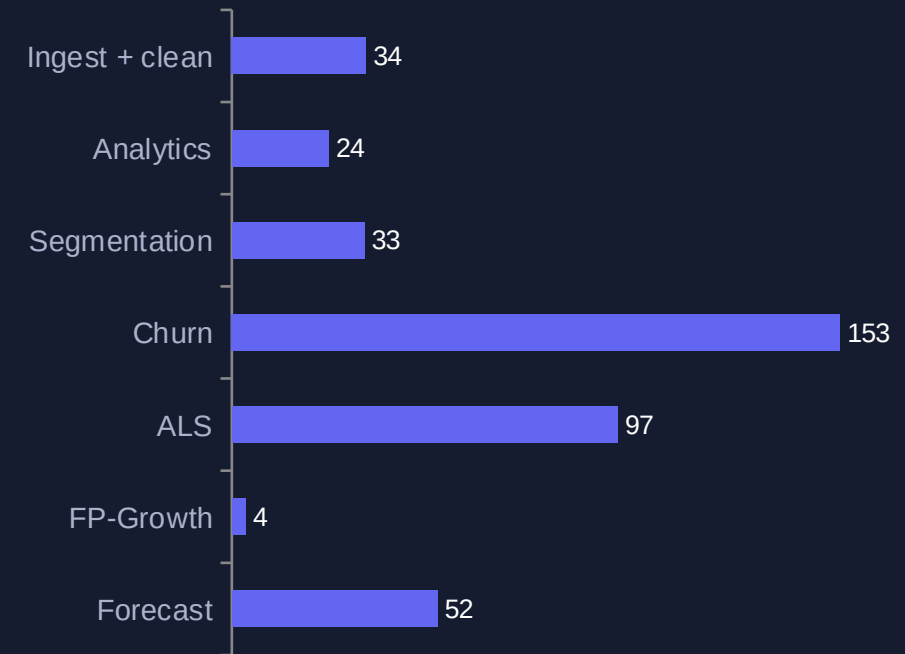
FastAPI + dashboard

Spark pipeline & data quality



Dataset	Raw	Clean	Fixed
Customers	10,033	10,000	33 dup · 107 cities
Orders	74,348	74,052	296 dup · 1,518 payments
Order items	1,45,368	1,45,160	208 invalid qty
Events	13,51,601	13,47,800	2,685 dup · 3,362 casing

Run on a 4-core laptop (medium scale)



Total: 6.7 min for ≈1.4 M records · every stage logged to MongoDB

Six machine-learning models (Spark MLlib)



Segmentation

RFM + K-Means

k = 5 · silhouette 0.375



Churn prediction

LR vs RF vs GBT

AUC 0.955 · precision 93.5%



Recommendations

Implicit-feedback ALS

HitRate@10 0.18 · 1.8× baseline



Market basket

FP-Growth

281 rules · lift up to 135×



Forecasting

LR trend + RF seasonality

**best daily MAPE 20.78% ·
weekly 6.78%**



Anomaly detection

Robust z-score (MAD)

**4/4 planted events · 0 false
alarms**

Customer segmentation: who are our customers?



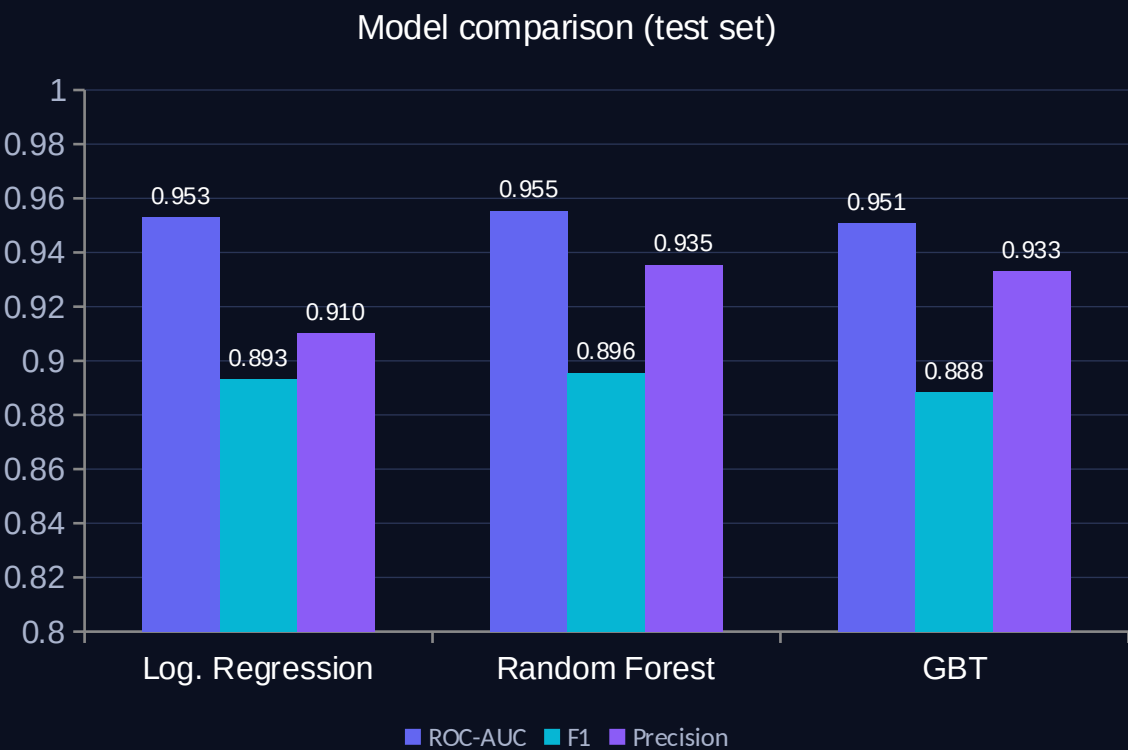
Segment	Customers	Avg orders	Revenue
Champions	1,394	30.1	71.58%
Can't Lose Them	1,541	7.9	15.21%
Loyal Customers	1,406	6.4	11.97%
Lost	1,223	1.7	0.85%
Hibernating	473	1.9	0.4%

23% of buyers → 72% of revenue

Champions dominate revenue; “Can’t Lose Them” (high value, silent ≈1 year) are the prime VIP win-back target. k chosen by silhouette over k = 3..8; clusters named automatically from centroids with a marketing playbook.

Churn prediction: who will stop buying?

Features: only data before cutoff (2026-04-02) → Label: no order in the next 90 days → 57.17% churned

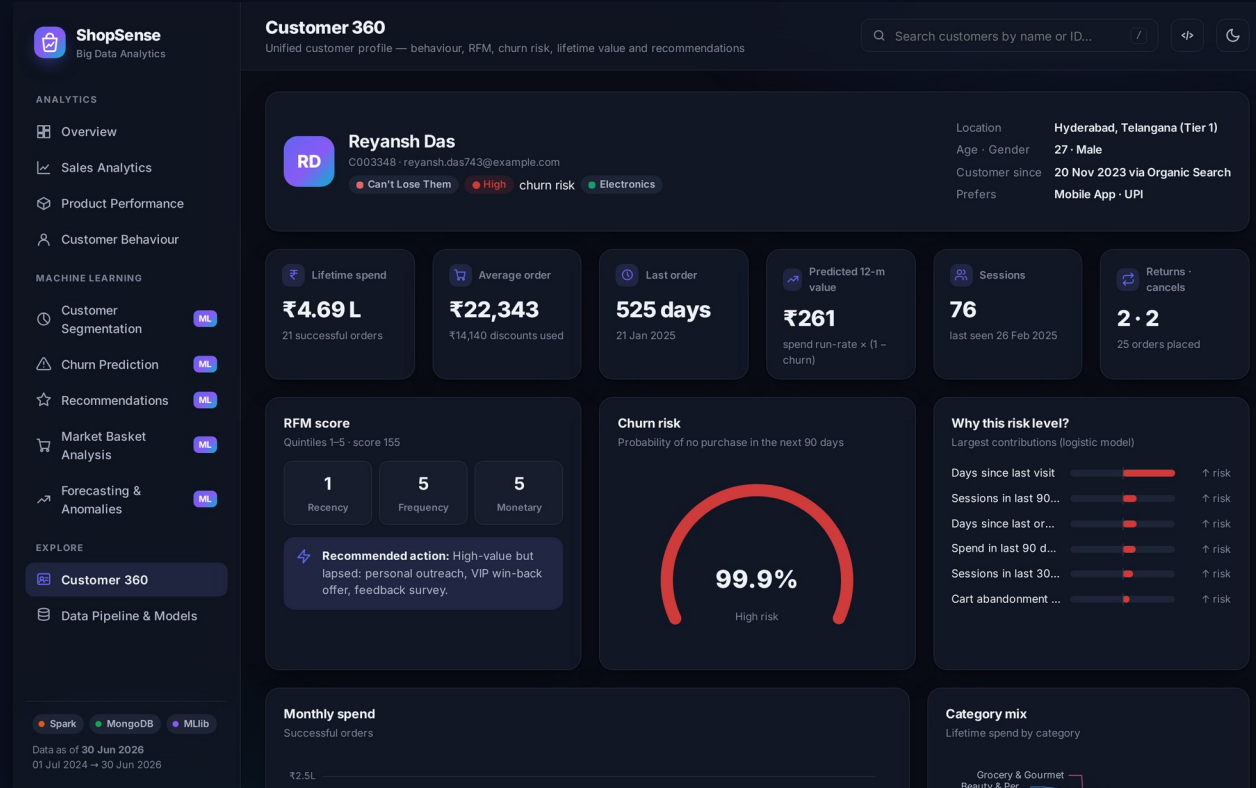
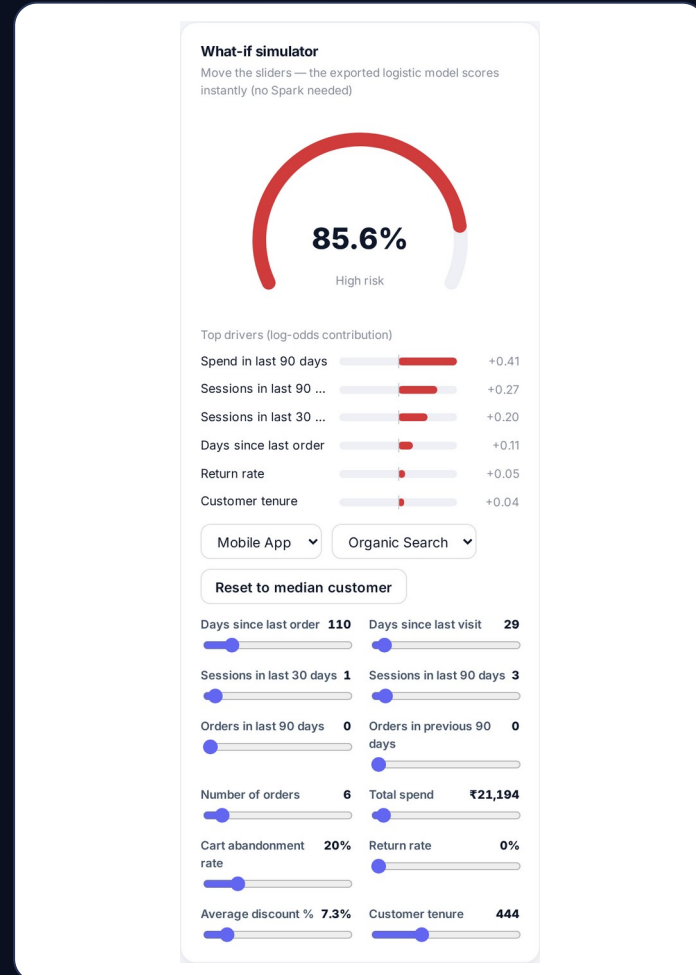


Top churn drivers (Random Forest)



2,956 high-risk customers • ₹8.37 Cr annual revenue at risk

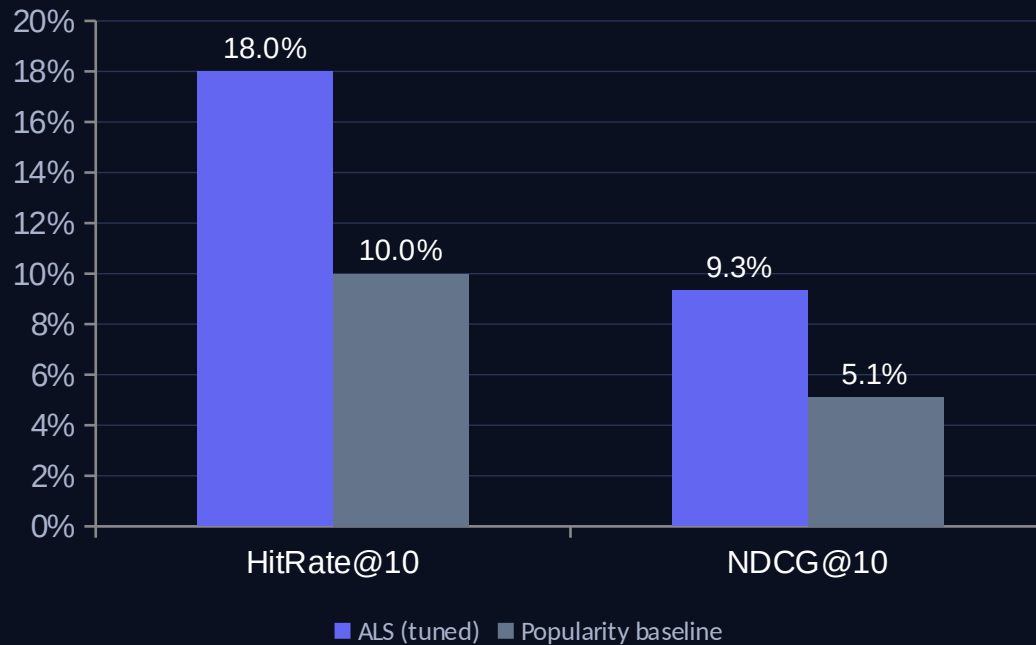
What-if simulator & Customer 360



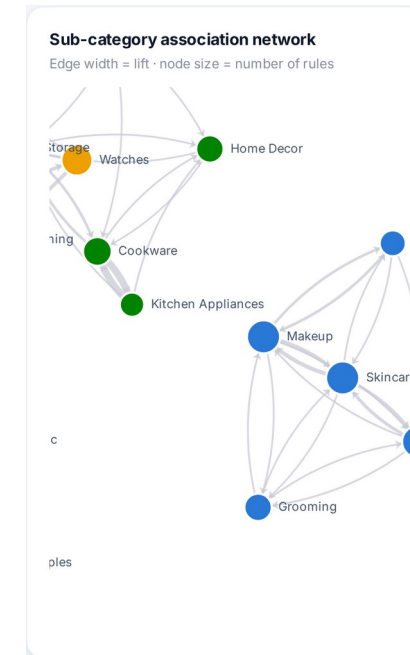
- Logistic model exported to MongoDB → API scores in ~1 ms without Spark
- Per-feature log-odds contributions explain every score
- Win-back list sorted by predicted 12-month value, exported as CSV

Recommendations & market-basket analysis

Leave-last-purchase-out evaluation

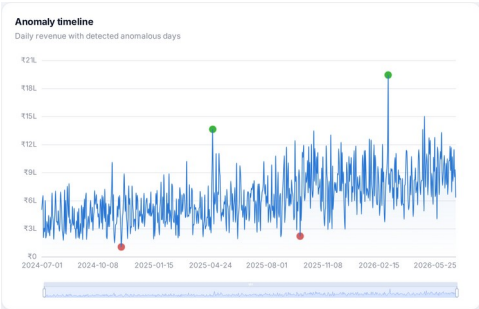
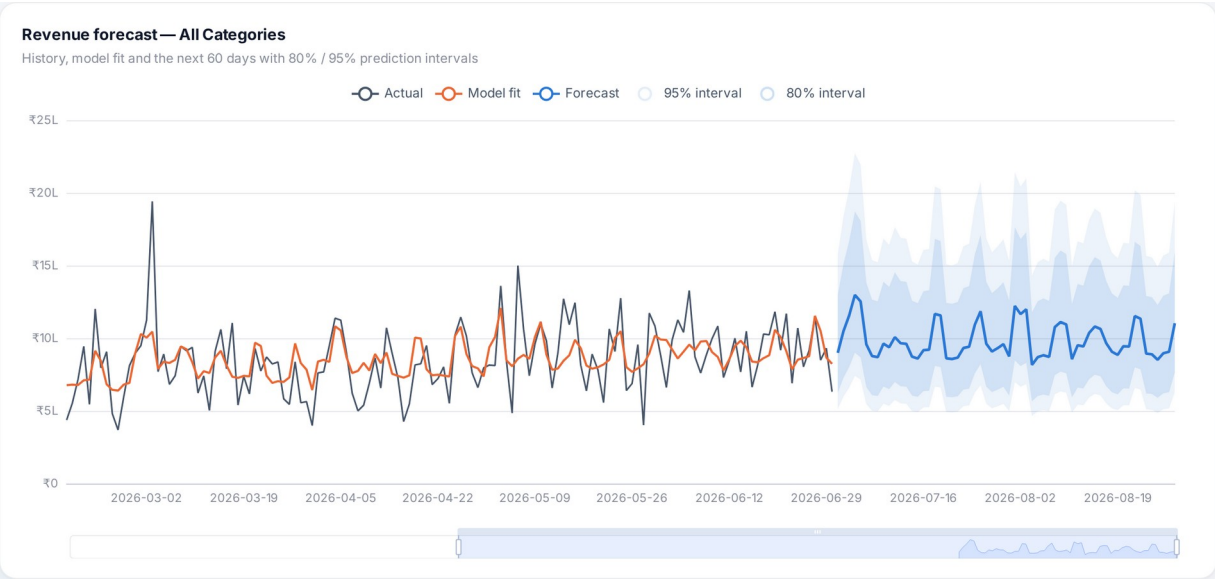


1.8× more hits than recommending best-sellers (rank 64, $\lambda=1$, $\alpha=5$, grid-searched)



- Cookware → Kitchen Appliances lift 10.5 · conf 41%
- Non-Fiction → Fiction lift 8.5 · conf 26%
- Toys → Baby Care lift 7.9 · conf 30%

Forecasting & anomaly detection

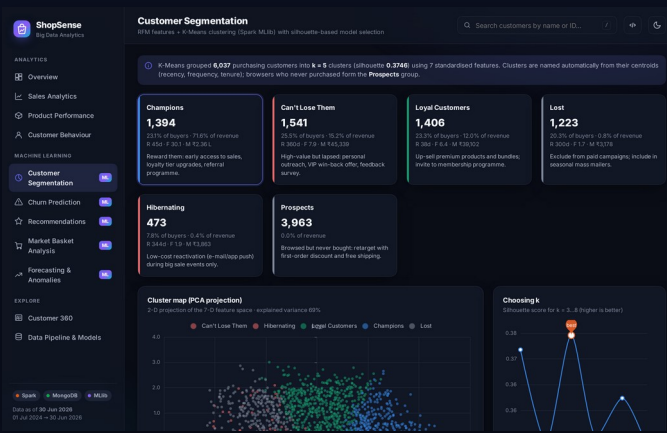
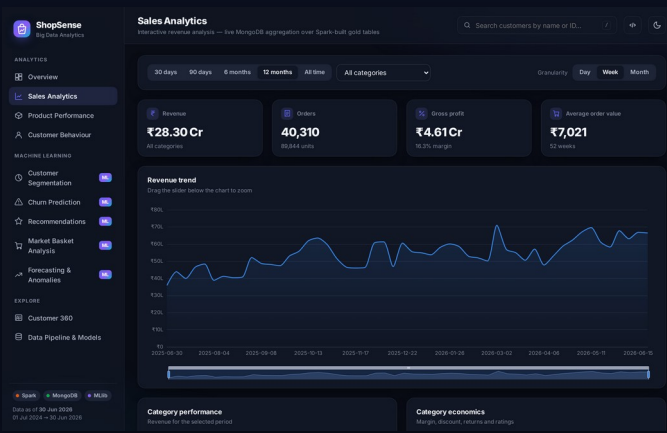


Model	Daily MAPE	Weekly
Hybrid (ours)	20.78%	6.78%
Calendar Linear Regression	22.72%	7.56%
Seasonal Naive	24.72%	6.31%
28-day Moving Average	21.91%	14.42%

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planted anomalies found, 0 false alarms
Gateway outage • viral flash sale • downtime • bulk order. Residual robust z-score ignores festivals that the model already expects.

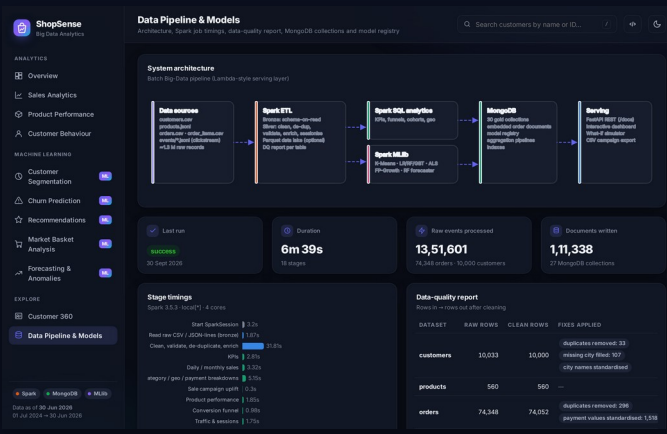
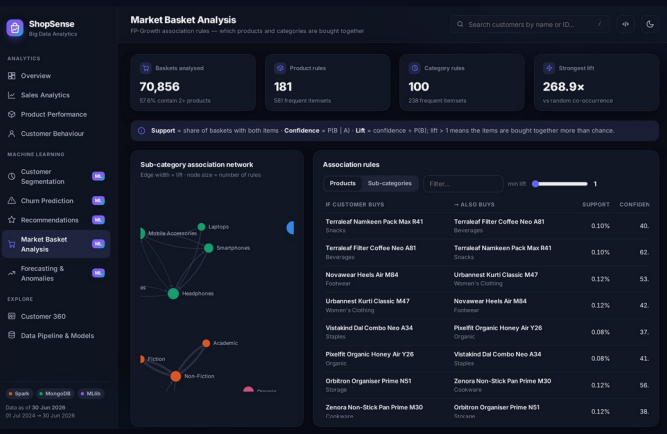
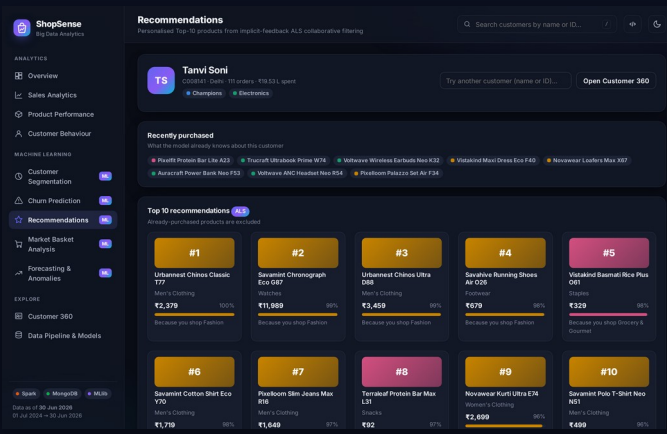
Live dashboard: 11 interactive pages



Sales analytics

Customer behaviour

Segmentation



Recommendations

Market basket

Pipeline & models

python run_dashboards.py → http://127.0.0.1:8000

Swagger API docs at /docs

Business insights discovered

+117%

orders during Big Festive Sale

but only +22–30% revenue: discounts eat the uplift

7.37%

Cash-on-Delivery cancellation rate

≈2× prepaid methods: nudge COD users to UPI

60%

of revenue from Electronics

at only 7% margin: cross-sell Fashion & Beauty (36–40%)

34.56%

cart abandonment

largest funnel drop is view → cart: recommendations matter

#1

churn signal = days since last visit

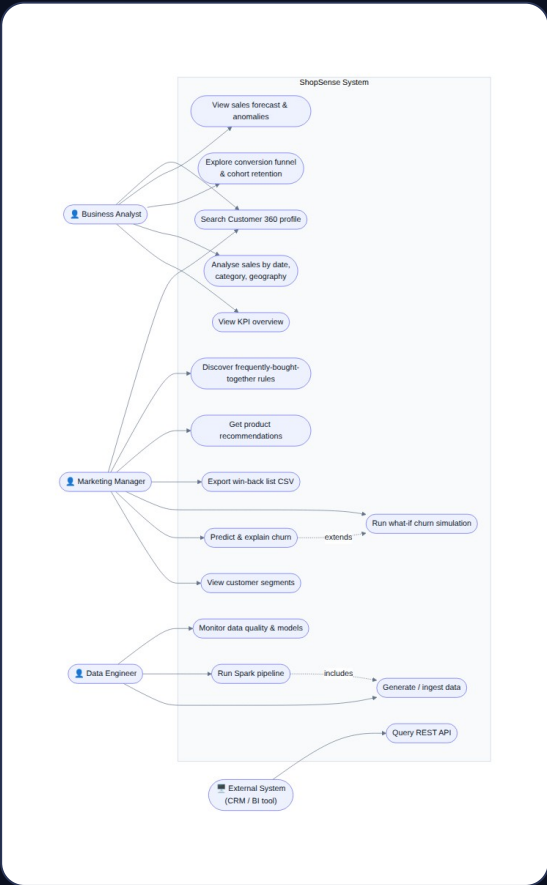
click-stream beats purchase history for early warning

₹8.37 Cr

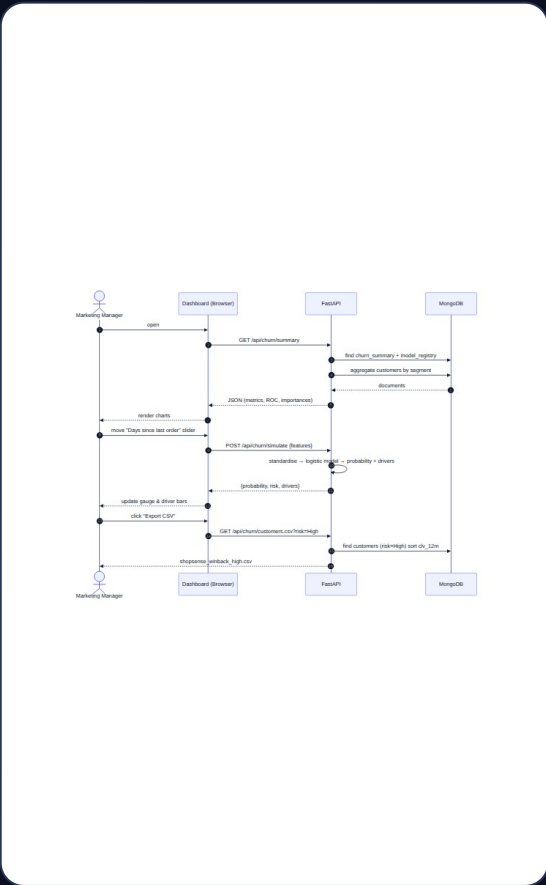
annual revenue at risk

2,956 high-risk customers exported for win-back

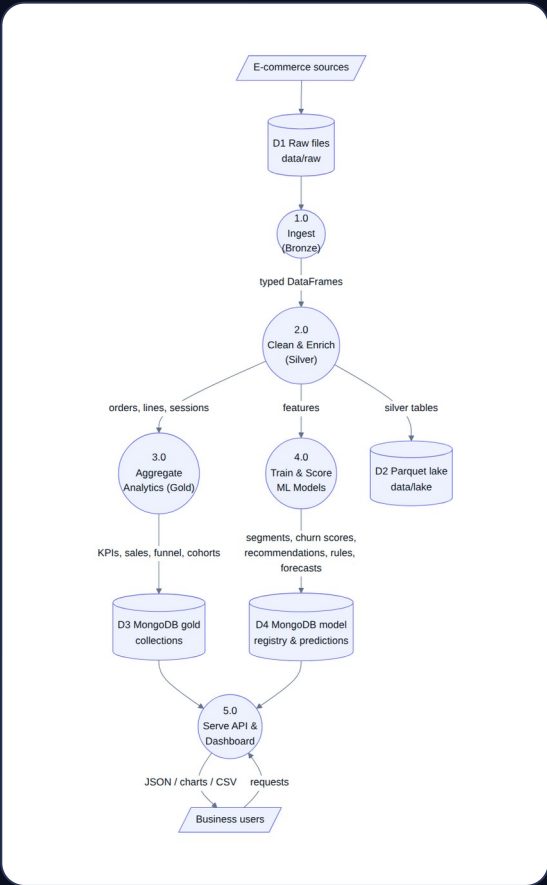
Design: UML & data-flow diagrams



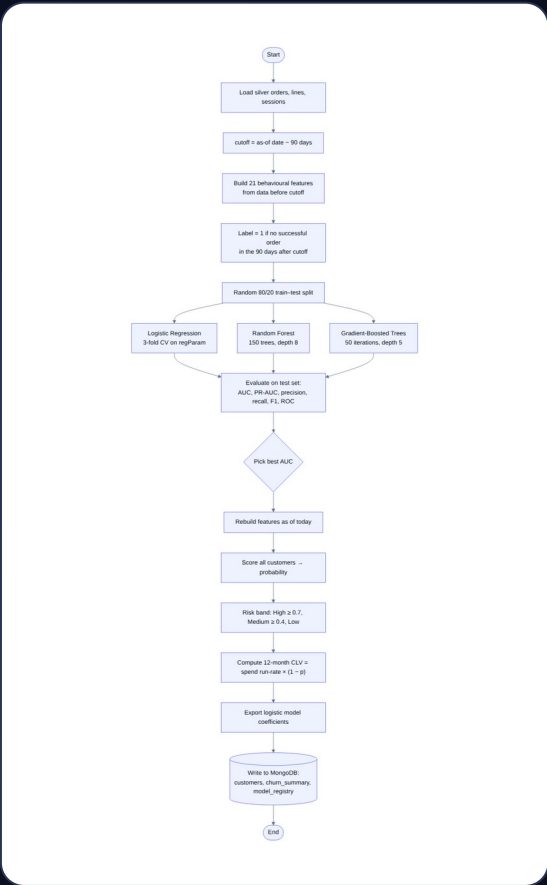
Use-case



Sequence (dashboard)



DFD level 1



Activity (churn)

Conclusion & future scope

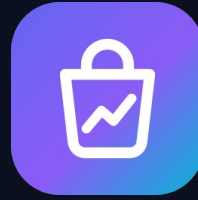
What we achieved

- End-to-end Spark → MongoDB → dashboard platform
- ≈1.4 M records processed in under 7 minutes on a laptop
- 6 MLlib models: churn AUC 0.955, ALS 1.8× baseline, 4/4 anomalies
- Explainable, actionable outputs: playbooks, simulator, CSV export
- 100% open source, reproducible, documented and tested

Future scope

- Real-time streaming with Kafka + Structured Streaming
- Cluster deployment (YARN/Kubernetes, Delta Lake, Atlas)
- Survival & uplift models, sequence-aware recommenders
- Review sentiment analytics with NLP
- MLOps: Airflow scheduling, MLflow tracking, drift monitoring
- Natural-language analytics assistant

Big Data tools from the curriculum, applied end-to-end to a real business problem.



Thank you!

Questions & discussion

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github.com/Interior-Gardener/BDA